

Fundamentos de Aprendizagem Automática

Learning as Compression & Information Foundations

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Course Introduction

Welcome to FAA

- **Curricular Unit:** Fundamentos de Aprendizagem Automática (41010)
- **Target:** Master's / Advanced Undergraduate in Informatics & Computer Engineering
- **Structure:** 3 hours per week (1h Theory/Presentation + 2h Hands-on Practice)
- **Instructor:** Mário Antunes (mario.antunes@ua.pt)
- **Philosophy:** From first principles to modern frontiers:
 - Foundations grounded in Information Theory and Compression
 - Comprehensive perspective across the learning paradigms
 - Hands-on mastery with Python and modern ML ecosystems

- **Discrete Evaluation Mode:**
 - **Theoretical Component (50%):**
 - ▶ Mid-Term Test: **25%** (Class 8, Week of 05-06 Nov 2026)
 - ▶ Final Test: **25%** (Class 14, Week of 17-18 Dec 2026)
 - **Practical Component (50%):**
 - ▶ Small Project 1: **25%** (Due Class 8)
 - ▶ Small Project 2: **25%** (Due Class 14)
- **Final Evaluation Mode:**
 - Final Exam: **50%**
 - Comprehensive Project: **50%**

What is Learning?

The Core Question

- What does it mean for a machine to *learn*?
- Classic definition (Tom Mitchell, 1997): > "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."
- Is learning just memorization?
 - A lookup table has perfect performance on seen data (E).
 - Does it generalize to unseen instances?

Generalization vs Memorization

- True learning requires **generalization**: identifying underlying regularities.
- The dilemma:
 - Complex models can memorize noise (**overfitting**).
 - Simple models may fail to capture true patterns (**underfitting**).
- How do we formalize the preference for simplicity?
 - Enter **Information Theory** and **Algorithmic Information Theory**.

Learning as Compression

The Compression View of Learning

- Fundamental thesis: **Learning is compression.**
- If a dataset can be compressed into a concise algorithm or model, regularities have been discovered.
- Random data (pure noise) cannot be compressed.
- If a hypothesis H explains dataset D with fewer bits than raw D , then H captures the regularities of the data distribution.

- The **Kolmogorov Complexity** $K(x)$ of a string x is the length of the shortest computer program that outputs x :

$$K(x) = \min_p \{|p| : U(p) = x\}$$

- $K(x)$ is non-computable, but serves as the theoretical ideal for compression and induction.
- Solomonoff Induction: Prediction by weighting all computable hypotheses inversely to their program lengths ($2^{-|p|}$).

Minimum Description Length (MDL)

- Practical computable formulation proposed by Jorma Rissanen (1978).
- To explain dataset D with hypothesis H , total code length is:

$$L(D, H) = L(H) + L(D \mid H)$$

- Where:
 - $L(H)$: Length in bits needed to describe the model / hypothesis (Model Complexity).
 - $L(D \mid H)$: Length in bits needed to describe the data given the model (Residual / Error).
- Minimizing $L(D, H)$ automatically balances goodness of fit with model complexity (**Occam's Razor**).

Connection to Modern Machine Learning

- Regularized loss in modern ML:

$$\mathcal{L}_{total}(\theta) = \underbrace{\mathcal{L}_{data}(D; \theta)}_{L(D|H)} + \lambda \underbrace{\Omega(\theta)}_{L(H)}$$

- Negative log-likelihood is Shannon code length:

$$-\log_2 P(D | \theta) \equiv L(D | H)$$

- Parameter regularization (L_1, L_2) corresponds to prior description length of weights.

Practical Session Preview

- Hands-on exploration in `practice/01_compression_md1.md`:
 1. Measuring entropy of sequences and texts.
 2. Polynomial regression under varying degrees.
 3. Measuring model complexity vs residual error (BIC, AIC, MDL).
 4. Visualizing how compression prevents overfitting.